

Connor Batsell

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EDUCATION

Rice University

Houston, TX

Bachelor of Science in Computer Science and Artificial Intelligence

May 2028

- **Relevant Coursework:** Introduction to Computer Organization, Introduction to Program Design, Computational Optimization for Artificial Intelligence, Linear Algebra
- **Clubs & Organizations:** Rice Computer Science Club, Machine Learning at Rice, HackRice

SKILLS

- **Languages:** Python, C/C++, Java, R, TypeScript, JavaScript, HTML, CSS, Excel
- **Frameworks & Libraries:** Spring Boot, Next.js, Node.js, React, LangChain, MyBatis, Tkinter, MCP SDK
- **Tools & Platforms:** MySQL, Docker, AWS ECS, Git, Linux, Ollama, REST APIs, OAuth 2.0, RAG

EXPERIENCE

Schreib Lab, Rice University

Houston, TX

Research Assistant

May 2026 – Present

- Building an ed-tech platform for Rice's COMP 312 (Programming Languages) under Dr. Rebecca Schreib, deployed to **200+ students** for step-by-step **Haskell** type-inference visualization.
- Architecting **Spring Boot** REST APIs with **MySQL** and **MyBatis** on a **TypeScript** frontend, supporting multi-user sessions, progress persistence, and instructor checkpoint management.
- Implementing a Hindley-Milner inference engine that traces unification and type substitution with annotated AST walkthroughs.

Trammell PC

Houston, TX

Software Engineering Intern

May – August 2025

- Proposed and led creation of an internal **MCP server** to automate client call logging and bulk document upload into the Clio legal practice management platform, eliminating **90%** of attorney manual data entry.
- Built on a **Node.js/TypeScript** backend using the official **MCP SDK** with OAuth 2.0 for both Clio API access and MCP client authentication, accessible to attorneys via Claude Desktop.
- Developed a **Next.js** admin dashboard for OAuth onboarding and audit log review; containerized with **Docker** and deployed on **AWS ECS** with autoscaling, supporting automated upload of **15,000+** client documents into Clio.

Zarvona Energy

Houston, TX

Software Engineering Intern

May – August 2024

- Developed an internal **Retrieval-Augmented Generation (RAG)** chatbot using **LangChain** and **Ollama**, integrating secure knowledge retrieval for **50+ engineers** and reducing support response time by **95%**.
- Developed a **Python** tool with a **Tkinter** GUI to evaluate gas-assisted plunger lift feasibility via pressure-volume calculations, flow-rate modeling, and well performance simulations on Excel field data.

PROJECTS

Dynamic Memory Allocator (C)

- Implemented malloc, free, and realloc in **C** with 8-byte alignment and boundary-tag coalescing, replacing an implicit free list baseline with segregated explicit free lists indexed by power-of-two size classes for near-constant-time allocation.
- Engineered split-on-allocate and immediate-coalesce-on-free policies, plus in-place expansion in realloc to avoid unnecessary memcpy calls and minimize heap fragmentation, achieving **90%+ space utilization** on trace-driven benchmarks.
- Wrote a verbose heap consistency checker validating block alignment, header/footer agreement, free-list membership, and coalescing invariants to surface pointer corruption during development.

FEAT: Automated Test-Case Generation System (Java, Python)

- Built typed **Python** object model for primitive and iterable types using generics, inheritance, and abstraction.
- Constructed end-to-end pipeline executing reference and buggy implementations via process-level invocation, aggregating failures across versions.
- Designed test selection via greedy set-cover algorithm, integrating recursive AST-based generators and JSON-driven configuration.